

Hellenic Complex Systems Laboratory

Calculator for Diagnostic Accuracy Measures

Technical Report XIII

Theodora Chatzimichail
2018

Calculator for Diagnostic Accuracy Measures

Theodora Chatzimichail ^a

^a tc@hcsf.com,

^a *Hellenic Complex Systems Laboratory*

Short Description of the Demonstration

This Demonstration calculates various diagnostic accuracy measures of a diagnostic test for a disease. This is done for different numbers of negative and positive test results in nondiseased and diseased populations. The measures calculated are the sensitivity, the specificity, the positive predictive value (PPV), the negative predictive value (NPV), the (diagnostic) odds ratio (OR), the likelihood ratio for a positive test result (LR+), and the likelihood ratio for a negative test result (LR-). The negative and positive test results of the nondiseased and diseased populations are selected using sliders.

Search Terms

sensitivity, specificity, diagnostic test, clinical accuracy, diagnostic accuracy, positive predictive value, negative predictive value, likelihood ratio, odds ratio

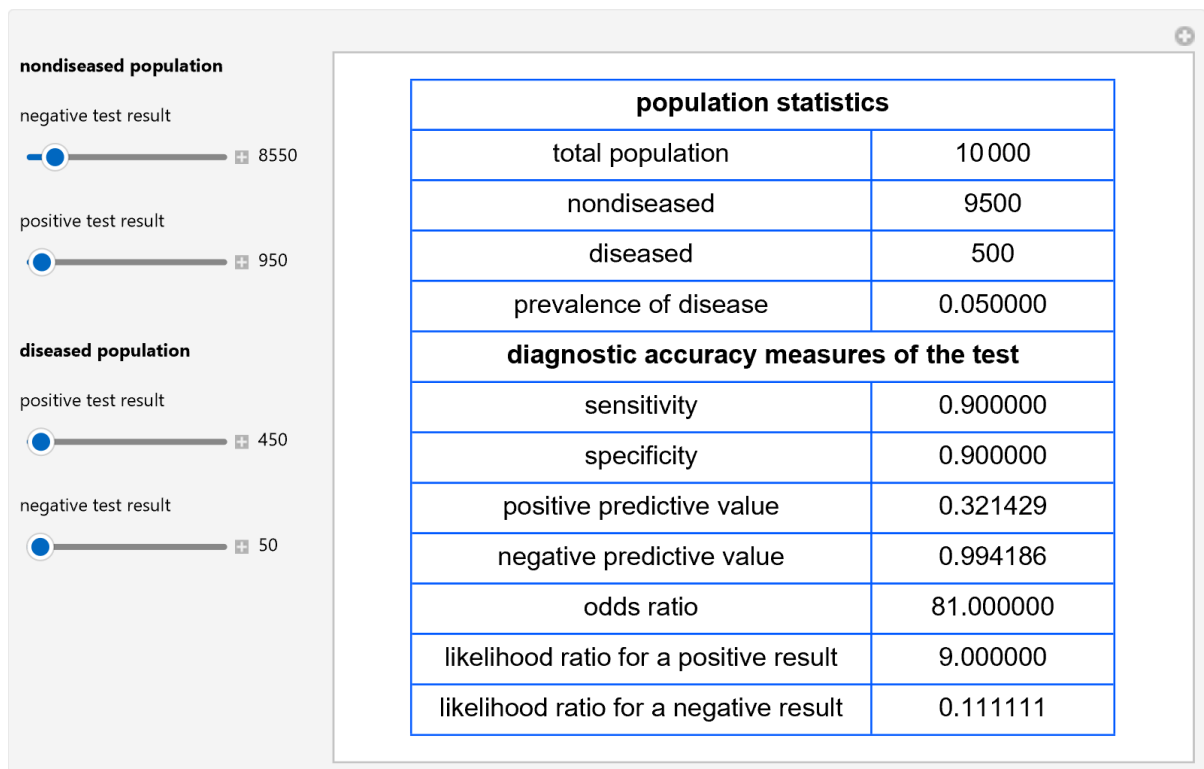


Figure 1: Population statistics and diagnostic accuracy measures of a diagnostic test, with the settings shown at the left.

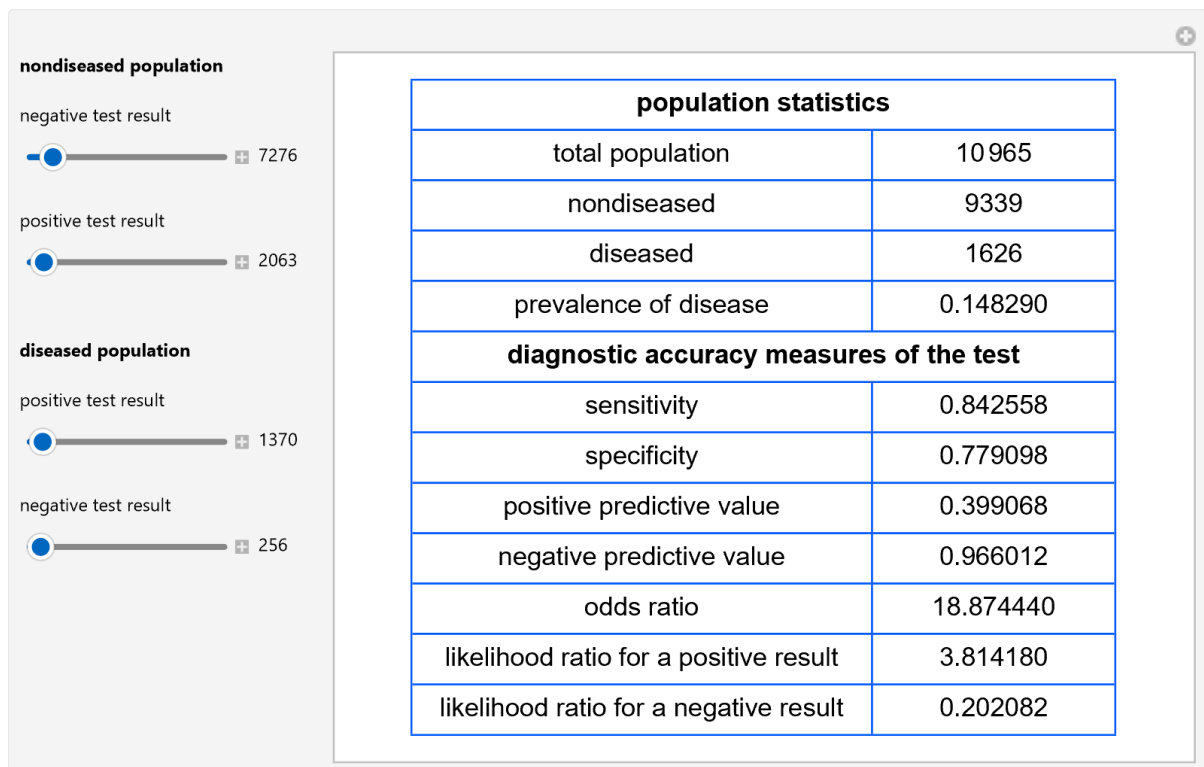


Figure 2: Population statistics and diagnostic accuracy measures of a diagnostic test, with the settings shown at the left.

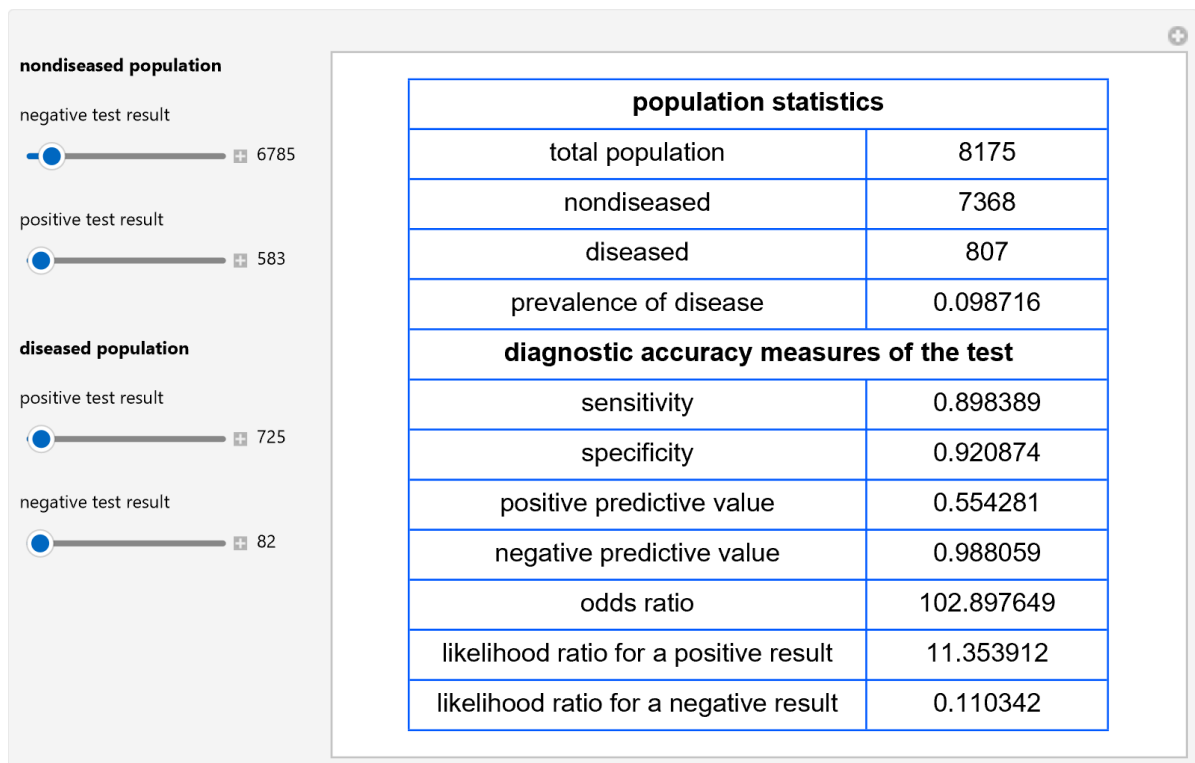


Figure 3: Population statistics and diagnostic accuracy measures of a diagnostic test, with the settings shown at the left.

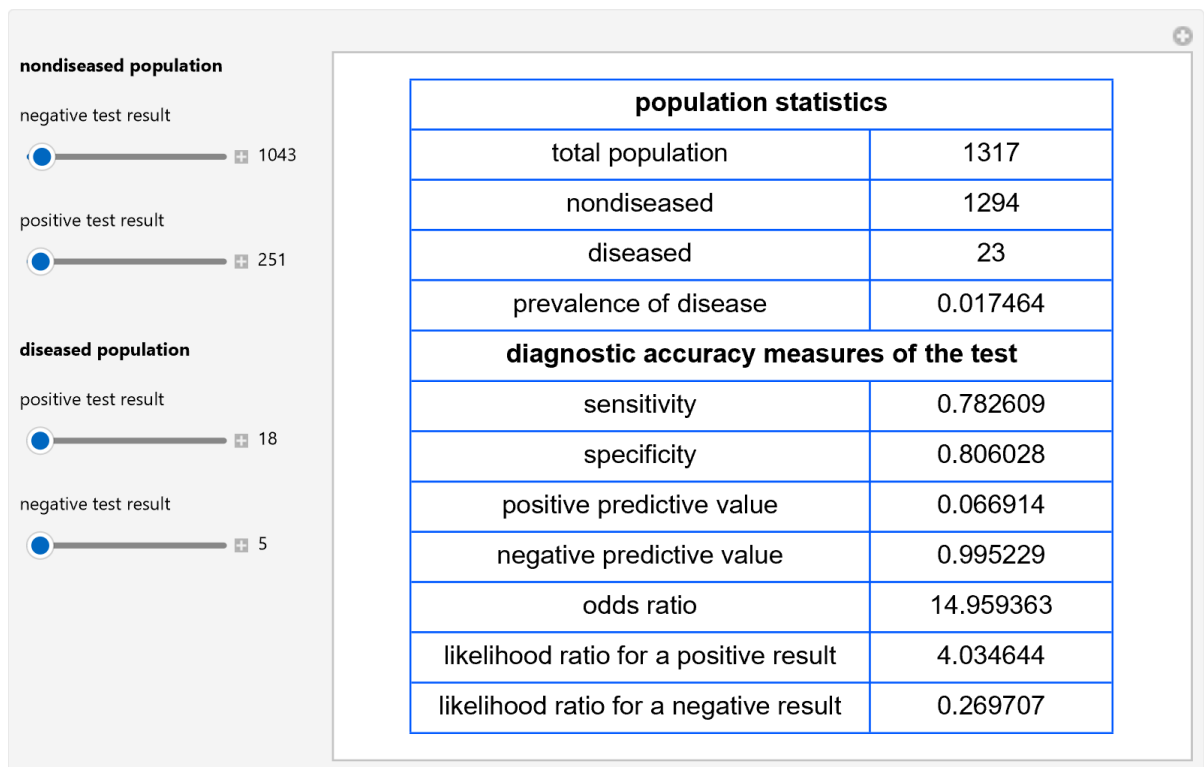


Figure 4: Population statistics and diagnostic accuracy measures of a diagnostic test, with the settings shown at the left.

Details

Sensitivity is the fraction of the diseased population with a positive test result, while specificity is the fraction of the nondiseased population with a negative test result. Positive predictive value is the fraction of the population with a positive test result that is diseased, while negative predictive value is the fraction of the population with a negative test result that is nondiseased. Prevalence of the disease is the ratio of the diseased population to the total (nondiseased and diseased) population. If we denote by Se the sensitivity, Sp the specificity, and v the prevalence, we have:

$$PPV = \frac{Se v}{Se v + (1 - Sp)(1 - v)}$$

$$NPV = \frac{Sp (1 - v)}{Sp (1 - v) + (1 - v) pr}$$

$$OR = \frac{\frac{Se}{1 - Se}}{\frac{1 - Sp}{Sp}}$$

$$LR+ = \frac{Se}{1 - Sp}$$

$$LR- = \frac{1 - Se}{Sp}$$

The Demonstration is appropriate as an educational tool in medical statistics and diagnostic test evaluation.

Demonstration

DOI

Source

Programming language: Wolfram Language

Availability: The updated source notebook is available at:

<https://www.hcsl.com/Tools/Demonstrations/CalculatorForDiagnosticAccuracyMeasures.nb>

First published in 2018 as part of the [Wolfram Demonstrations Project](https://demonstrations.wolfram.com/):

<https://demonstrations.wolfram.com/CalculatorForDiagnosticAccuracyMeasures/>

Software Requirements

Operating systems: Microsoft Windows, Linux, Apple macOS and iOS

Other software requirements: Wolfram Player®, freely available at: <https://www.wolfram.com/player/> or Wolfram Mathematica®.

System Requirements

Processor: x86-64 compatible CPU.

System memory (RAM): 4 GB+ recommended.

Citation

Chatzimichail T. *Calculator for Diagnostic Accuracy Measures*. Ver. 1.1.1. Hellenic Complex Systems Laboratory; 2026.

License

[Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License](https://creativecommons.org/licenses/by-nc-sa/4.0/).

Technical Report

DOI

Citation

Chatzimichail T. *Calculator for Diagnostic Accuracy Measures*. Technical Report XIII. Hellenic Complex Systems Laboratory; 2018.

License

[Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License](https://creativecommons.org/licenses/by-nc-sa/4.0/).

First Published: April 25, 2018

Revised: May 23, 2026